Pennysia: An Open Liquidity Network

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Abstract

Pennysia is an open liquidity network, a single exchange layer where every source of liquidity, wherever it sits, competes for the user's order. Decentralized exchanges, aggregators, intent solvers, market makers, perpetual and prediction protocols, vaults, fiat ramps, and licensed venues all answer the same auction. The best executable price wins. The user signs once. The network routes, executes, and verifies the fill. For builders, a venue connects once and competes for every order from day one.

The result blurs the line between centralized and decentralized exchange, pairing the performance, liquidity, and experience of the former with the custody of the latter. No seed phrases, no gas tokens, no chain selection. Fiat enters and exits on card and bank rails. The experience reads like a centralized exchange. The guarantees read like a DEX.

Liquidity is organized into two lanes: a **permissionless lane** of open venues, onchain and offchain, open to everyone, and a **regulated lane** of gated liquidity (centralized market makers, licensed venues, compliant pools, fiat ramps) unlocked by optional verification. Compliance becomes a routing dimension, not a gate.

An experimental deployment on Ethereum proved the model: 30 venues competing in parallel per swap, atomic in-transaction fallback, and a settlement layer that enforces signed terms on every order. Pennysia v1, the Open Liquidity Network, generalizes that proof to every chain and every service class, a single interface for the world's liquidity.

This paper describes Pennysia v1, the Open Liquidity Network, as designed. Design decisions open at publication are noted where they occur; the Ethereum deployment documented in Appendix A demonstrates the core mechanisms in production.

1 Introduction

Crypto liquidity is fragmented along two axes at once [13], and every exchange today serves only a slice of the result.

The first axis is **chains**. Liquidity is spread across dozens of networks (Ethereum and its L2s, Solana, BNB Chain, and a growing set of new execution environments), each with its own DEXs, aggregators, market makers, and intent protocols. No single venue wins on every pair, size, and moment. A trader who chases prices across chains inherits bridge risk [15], gas-token juggling, and a different mental model per network. Chains behave like disconnected internets. A service lives where it deployed, and reaching it elsewhere means switching providers. Even the strongest onchain venues remain single-domain: excellent execution on their own chains and instruments, with the rest of the market outside reach.

The second axis is the **protocol boundary**: the split between *open* and *closed* protocols. Open protocols (DEXs, intent networks, onchain markets and vaults) are permissionless, transparent, and self-settling, but shallow on size [16]. Closed protocols (centralized books, market makers, OTC desks) hold deeper, tighter liquidity behind accounts, KYC, and custodial terms. Traders choose between two products and two trust models, with no common interface. The deepest quote in the world is invisible to an open-protocol trader.

Between the trader and both axes sits **the UX wall**. Self-custody is the founding premise of this space: Bitcoin was built so that holding and moving value requires no trusted intermediary [10]. Its cost has never been ideological; it is operational. Seed phrases gate the first ten minutes. Gas requires a native token on every chain. Chain selection, bridging, and per-venue approvals turn one trade into a checklist, and the endpoints (buying in with a card, cashing out to a bank account) still route through custodial platforms. Each step is a point where users exit to a centralized exchange's account login. The market's trade-off today: **easy to use, or yours.** Centralized exchanges offer the experience; decentralized venues offer the custody. No venue offers both. Figure 1 lays out both journeys.

Three enablers make this the moment to build it. Intent-based execution [3] turned "user signs, someone else transacts" into production infrastructure, the mechanical basis for gasless, chainless trades. Embedded wallets matured [4] to where a Web2 login (email, Google, Apple) provisions real EVM and Solana wallets, giving Web2 onboarding with

	On-ramp	*Wallet*	*Activity*	*Off-ramp*
DeFi	1. open a CEX or card-ramp account 2. pass KYC either way 3. buy the asset 4. wait out the withdrawal lock	5. install a wallet 6. write down the seed phrase 7. acquire the gas token 8. withdraw to the right chain	9. pick venue and chain 10. approve token (tx + gas) 11. swap (gas) 12. bridge if needed (gas + wait)	13. sell into a stablecoin 14. send back to a CEX 15. sell for fiat 16. bank withdrawal
CeFi	1. open account 2. pass KYC 3. deposit by card or bank	4. balance and keys held by the venue	5. trade on the venue's internal ledger	6. bank withdrawal

Figure 1: Illustrative fiat-in to fiat-out journeys. Self-custody: ∼16 steps across three platforms, with seed custody, gas, and chain choice at every boundary; card funding still means a KYC'd ramp at a markup. Custodial: ∼6 steps, because the venue holds the keys. Design target: self-custody at the custodial step count.

user-side keys. And venue proliferation across both liquidity kinds makes comparison a routing problem, not a shopping problem. Figure 2 charts the migration already underway.

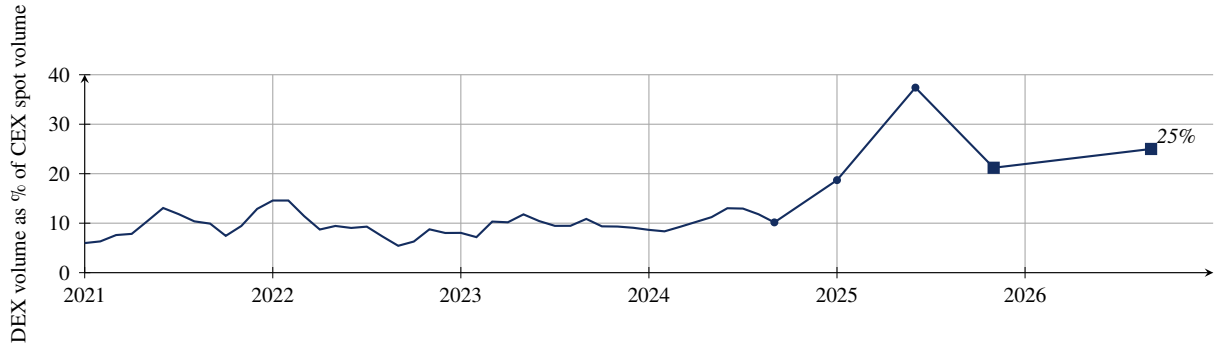


Figure 2: DEX spot volume as a share of CEX spot volume, 2021–2026: 6% to ∼25%. Line and circles: CoinGecko [5] (monthly through Sep. 2024; landmark months after). Square: The Block's wider top-30-DEX series, Sep. 2026 [12], indicative not strictly comparable. The migration is real but incomplete; the UX wall is the remaining ceiling.

Pennysia v1, the Open Liquidity Network (OLN), generalizes the Ethereum prototype's proof (Appendix A) along both fragmentation axes: a unified layer spanning **every chain** and **every liquidity service** (spot today; perpetuals, prediction markets, and yield protocols next), wrapped in an account that removes the UX wall. Pennysia implements none of these services; it operates the network that connects to all of them. The end state is a superset of the centralized exchange: every essential feature (one login, fiat rails, deepest liquidity, more tokens, fast settlement) with custody that never leaves the user's keys. For builders the network is distribution: a new protocol connects once and competes for every order.

2 Design Principles

Six commitments govern the design. Where an implementation decision remains open, the governing principle is stated in its place.

- **Liquidity is venue-agnostic.** No venue is special. Every liquidity source and service protocol reduces to the same abstraction, *a source that returns an executable price*. Pennysia builds no competing venue, pool, or protocol of its own. It operates the network that connects to all of them.
- **Non-custodial by construction.** The network never holds user funds. No pooled balances, no deposit accounts, no withdrawal queue. If the network disappeared tomorrow, every asset would remain in users' wallets.
- **Signature-first UX.** Users sign. The network transacts. A trade is one review and one signature. Gas, chain selection, submission, and confirmation are the network's problem.
- **Competition decides, not trust.** Quotes rank by **net executable output** (what lands in the wallet after gas, fees,

and fill probability), not headline price. No venue buys placement. Listing is free, rank is earned per order.

- **Compliance is a lane, not a gate.** Verification is opt-in and rewarded with liquidity, never required as the price of entry.
- **Honest fees.** Costs live inside the quote, never after it. The network earns when execution beats the quote, not when it does not. There is no native token, and access is not gated by holdings.

3 System Overview

Four layers, each with one job (Figure 3). Users interact only with the top. Venues integrate only with the middle.

Experience	one account, one balance, one ticket; chain & gas hidden, status to finality
Identity	login → non-custodial wallets (embedded or external); optional KYC
Quote market	parallel fanout → ranked net output; permissionless lane | regulated lane
Execution	signed order → best venue → verified fill; Settlement, intents, venue rails

Figure 3: Four layers, each with one job. Users interact only with the top; venues integrate only with the quote market and execution layers.

Identity provisions non-custodial embedded wallets per chain family from a Web2 login (email, Google, Apple); external wallets connect for the same surface (Section 4). The **quote market** fans every order out in parallel to all venues in the user's enabled lanes and normalizes and ranks the quotes (Section 5). The **execution fabric** carries a signed order to settlement over three rails (atomic routes, intent rails, venue rails) under one set of guarantees (Section 6). The **experience** presents a unified balance, one trade ticket, and one status line, with chains visible only on request; portfolio and verification state also live here (Section 8).

Every order walks the same loop (quote, sign, execute, verify) regardless of which rail carries it. Figure 4 draws the boundary each participant sees.

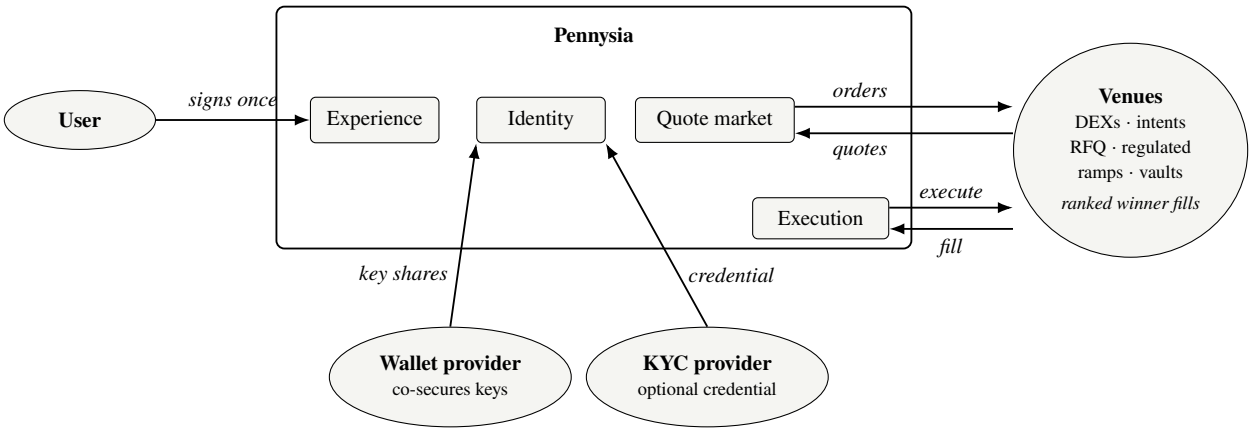


Figure 4: How each participant connects. The user touches only Experience; wallet and KYC providers touch only Identity; venues see only the quote market and execution layers, and only the ranked winner receives the signed order. Pennysia signs nothing and holds nothing.

4 The Account

The account is where the CEX/DEX line is deliberately blurred. Signup reads like a centralized exchange. Custody is a self-custody wallet.

4.1 Onboarding and custody

A user creates an account with a Web2 login (email, Google, or Apple). No seed phrase, no browser extension. Provisioning creates **embedded wallets**, one per supported chain family (EVM and Solana at launch). The wallets are real onchain addresses belonging to the user. Assets sent to them are theirs, and every order signed is a standard wallet signature. A Pennysia account is a key management scheme with a login attached, not a database balance. Table 1 draws the boundary.

Table 1: Authority boundaries: what each actor can and cannot do.

Actor	Can do	Cannot do
User	Sign orders and transactions; export and self-manage keys	*none*
Pennysia	Verify sessions, assemble quotes, submit signed orders, display state	Move user funds; sign for the user; recover wallets
Wallet provider	Co-secure key material (device/enclave shares); assist user-driven recovery	Unilaterally spend

The load-bearing property is that session verification exists to assemble quotes and operate the account, never to move funds. Every value-moving operation traces to a signature produced in the user's session.

4.2 Gas and wallet choice

Because execution is signature-first (Section 6), gas is paid by fillers, venues, or relayers and priced into the quote. A user holding only USDC can trade it for output on any chain without acquiring a native token. Where a user's own onchain action is unavoidable (e.g., a first approval), the network sponsors or abstracts the gas rather than assigning a chain-specific errand.

Embedded wallets are the onboarding path, not the only path. An external wallet (extension, WalletConnect, hardware) reaches the identical quote market and execution fabric. The account then acts as a unified view and session.

4.3 In and out

On- and off-ramps are a service class, not a feature Pennysia builds. Ramp providers (cards, bank transfers, Apple Pay, Google Pay) return executable quotes for fiat ↔ token conversion and compete in the same auction (Section 5). Fiat rails are regulated by nature, so they sit in the regulated lane. The optional verification credential (Section 8) unlocks them. Cash in, cash out. Never a custodial balance in between.

Recovery is the wallet provider's domain. Re-authentication restores key shares without Pennysia involvement. Embedded wallets are ordinary onchain addresses and orders are standard signed payloads, so a user who leaves keeps full control. No exit cost by design.

Open design decision. Signature model under embedded wallets: plain EOA + typed-data intents (EIP-712-class [7]) vs. smart accounts (ERC-4337 [1], EIP-7702 [2]) with session keys and programmable recovery. The paper describes behavior both satisfy (signature-first, gasless, non-custodial) without committing the primitive. Resolve before spec detail lands in Appendix B.

5 The Quote Market

Liquidity today sits on protocols the way accommodation sits on hotels. Thousands of providers serve the same essential need, at different prices and terms, and a user should no more browse every venue than a traveler should browse every hotel's website. The quote market is the routing layer. It turns a fragmented liquidity landscape into a single price. Every order is an auction, every eligible venue sees the same request simultaneously, and the best *executable* answer wins [14].

5.1 The auction

A trade request (sell X, get Y) broadcasts to every venue in the user's enabled lanes. Formally, for user u each venue v in the eligible set $E(u)$ returns a quote (y_v, c_v, p_v): gross output, all-in cost (gas, venue and protocol fees), and fill probability, with expected settlement latency folded into the ranking. Quotes answer within a response window τ set by

the market. A venue that misses the window does not rank, so every quote in the auction is a live commitment rather than indicative pricing. The market selects the venue maximizing net executable output:

$$E(u) \;=\; \mathcal{V}_{\text{perm}} \;\cup\; \mathbf{1}[\text{verified}(u)] \cdot \mathcal{V}_{\text{reg}}, \qquad v^*(o) \;=\; \underset{v \in E(u)}{\arg\max} \; \underbrace{p_v \cdot (y_v - c_v)}_{\text{net output}} \tag{1}$$

A quote 0.3% better that fails 5% of the time loses to a slightly worse quote that settles. This ranking already runs on the Ethereum prototype; the OLN applies it identically across chains and venue classes.

5.2 Two axes of liquidity

Liquidity fragments along two independent axes: **permission** (who is allowed to serve a given user) and **location** (onchain protocols and contracts, or offchain books and inventory). The axes are orthogonal. A walletless provider quotes permissionlessly from offchain inventory, while a compliant pool is permissioned yet holds its liquidity onchain.

The permission axis defines the user's *lane*. A permissionless venue is an open city, a regulated venue a licensed jurisdiction. The **permissionless lane** is open to every user: every open liquidity source (DEXs, aggregators, intent networks, RFQ makers, walletless providers), carrying the full prototype venue set. The **regulated lane** unlocks through optional verification (Section 8): centralized market makers, OTC desks, CEX-linked liquidity, and fiat ramps, where depth on large size tends to live.

The location axis predicts the execution rail (Section 6). Atomic and intent rails settle onchain liquidity, and venue rails bridge offchain books to onchain delivery.

The lanes share one auction and one ranking. A regulated venue enters the same competition and wins by the same rule.

5.3 Venue and service classes

Figure 5 maps the landscape onto both axes.

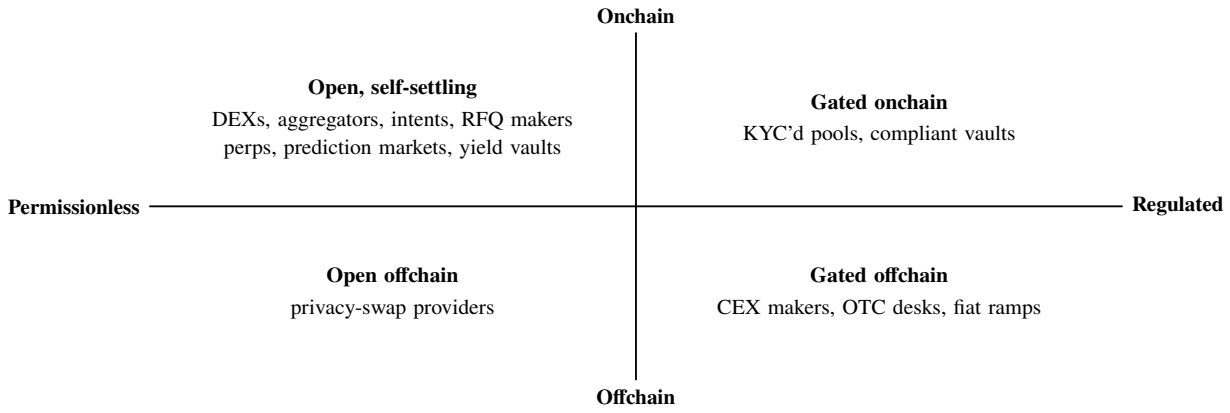


Figure 5: Two axes of liquidity. The horizontal axis is permission, the user's lane; the vertical axis is where the liquidity lives. Every class enters the same auction behind the same signed-terms gate.

The classes extend beyond swaps. Any service returning an executable price for a position (a perpetual, a market share, a vault deposit, a fiat conversion) enters the same auction under the same abstraction. Coverage compounds with every connection. Each chain adds its venue set, each venue extends the token surface, and a token liquid anywhere becomes tradable everywhere.

Joining differs by lane. The permissionless lane is free and self-serve. Any venue that returns executable quotes against the network's adapter interface joins the auction, the model that grew the prototype to 30 venues. The regulated lane adds a compliance onboarding covering licenses, jurisdictions, and settlement rails. Once connected, a venue competes for every order from day one, ranked on output alone.

5.4 Chainless quoting

A quote request names assets and intent, not a chain. Chains are plumbing in this model, an internet provider for the services deployed on them, relevant to routing but irrelevant to the requester. The routing layer resolves where the fill

happens (same-chain, cross-chain via fillers, or a regulated venue), and the user reviews net output and rail type, never a chain diagram. Chainless is the property of this liquidity network.

6 Execution

Execution turns a signed order into settled tokens. The user's part ends at the signature.

6.1 Signature-first order flow

The user reviews a quote stating the worst acceptable output (**WAP**, the minimum received, net of all fees) and signs an order, a typed-data payload binding input, output, floor, deadline, and recipient (Appendix B). No transaction, no gas prompt, no chain selection. The network derives the full execution plan and carries it to completion.

6.2 The three rails

Orders settle over three rails (Figure 6).

Rail 1: Atomic routes (same chain). A Settlement-class router pulls the user's input once (signed allowance authorization), attempts ranked venues inside one transaction, first success wins. All routes fail → full revert, with no partial fills, no stranded funds, no second signature. This is the prototype's production model (Appendix A), replicated per EVM chain.

Rail 2: Intent rails (cross-chain / solver fills). The signed order enters an intent settlement network. A filler commits to the quoted output, pays all gas, and delivers on the destination chain. User exposure is bounded by the signed terms: floor, deadline, recipient. Unfilled intents expire. An order never moves funds without delivery.

Rail 3: Venue rails (regulated lane). Centralized venues settle via escrow contract (atomic onchain exchange [9]) or deposit-address rails, the privacy-swap pattern the prototype already runs, where input goes to a venue address and output arrives asynchronously. Both bounded by the signed terms and reconciled on completion. The counterparty model is disclosed per quote (Section 7.4).

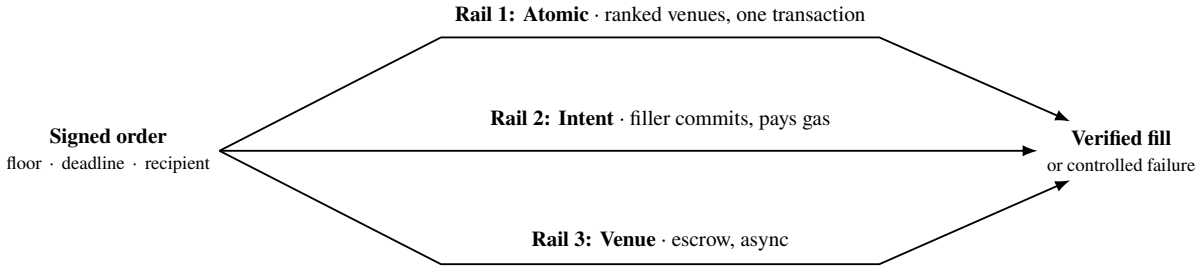


Figure 6: Three rails, one order format, one set of guarantees. The rails differ; the signed terms do not.

6.3 Guarantees and failure

One rule across rails: **funds move only under signed terms; a failed route returns control, not loss.** Formally, for every order o signed by user u,

$$\text{moved}(u, o) \implies \text{delivered}(u, o) \geq \text{WAP}(o) \ \wedge \ t \leq \text{deadline}(o) \tag{2}$$

Atomic routes revert wholesale, intents expire unpaid, and venue-rail fills resolve to delivered or refunded. Where a route set exists, the network attempts the next ranked venue before surfacing failure. Table 2 summarizes the full contract.

Latency is part of the contract. Same-chain fills settle within a block, while slower rails carry a bounded ETA priced into the quote, targeting centralized-exchange responsiveness on the fast rail. The surface stays minimal. Pick input and output, see one net number and an ETA, sign once. The rail underneath is visible on demand, never required.

Open design decision. Atomic routes as one Settlement-family contract per chain vs. a canonical intent-native executor shared across chains. Both preserve the semantics; the choice is engineering and audit scoping.

Table 2: User-facing guarantees and the mechanism behind each.

Guarantee	Mechanism
Worst-case output known at signing	WAP embedded in the order; enforced per rail
No gas token required	Executor pays gas, priced into quote; sponsored where unavoidable
No chain management	Rail and destination are the network's problem
Bounded failure	Revert, expiry, or refund; never partial loss
Disclosed settlement time	Single-block same-chain; latency priced into the quote as ETA
One signature	Review → sign → status to finality

7 Security Model

The prototype's model was double-layered, venue protections plus Pennysia's signed-terms gate. The OLN adds a third: the venue, the execution layer, and the account, each with a distinct job.

7.1 Venue protections

Every venue keeps its own security properties (audited contracts, solver bonding, maker collateral, exchange controls) exactly as if the user traded on its frontend. Pennysia adds on top. It never treats a venue as trusted.

7.2 Enforced execution

The network enforces the signed order mechanically. The counterparty trust that remains is bounded and disclosed (Section 7.4).

- **Signed bounded terms.** The signature caps the worst case: WAP floor, deadline, recipient. No venue can expand it.
- **Validation before execution.** Every route and quote is checked regardless of source; headline price authorizes nothing. Execution-order integrity on public chains is adversarial by default [6]; the network validates rather than assumes ordering.
- **Unified approvals.** Atomic routes use one Settlement approval instead of per-venue spenders.
- **Verified fills.** A fill is reconciled against the signed terms (sender, recipient, input, minimum output) before it is recorded as settled. Overquotes and mismatches are rejected, not absorbed.
- **Atomic fallback.** A failed venue is swapped for the next best inside the same transaction where the route set allows.

7.3 Account and key security

The custody table in Section 4 is the security claim. Pennysia's servers are never signers, sessions authorize only quote assembly and account operations, and the network holds no pooled user assets to attack.

7.4 Honest risk accounting

The residual risks, stated plainly:

- **Cross-chain latency.** Intent fills are asynchronous; the market can move between signing and delivery. The risk is an expired order, not a worse fill, but expiry is a cost.
- **Solver/filler dependence.** Intent rails inherit solver liveness. Multiple filler networks plus atomic fallback mitigate, not eliminate.
- **Regulated-lane counterparty exposure.** Venue rails extend bounded trust to a centralized venue for the settlement window, a different trust profile than an atomic swap, disclosed per quote, which is why the lane is opt-in.
- **Approval concentration.** Unified approval concentrates exposure in the Settlement contract, hence open source plus four audit toolchains (Appendix A) as the bar for every deployment.
- **Embedded-wallet trust model.** Keys are co-secured with the wallet provider (threshold/MPC-class custody [8]). External wallets remain first-class for users preferring a different model.

New contracts follow the prototype's bar: open source, unit/fuzz/invariant coverage, symbolic execution, coverage-guided fuzzing, and formal verification where tractable. Deployment artifacts and audits publish with each chain rollout.

The compounding benefit is uniformity. Accessing protocols directly means inheriting each one's risk surface. Routing through the network collapses that to a single security posture: one interface, one signature, one guarantee set, one point of contact, with the signed-terms gate between the user and every provider's individual risk.

8 Compliance Posture

The OLN's answer to regulation is structural. **Compliance is a routing dimension.** The permissionless lane is the default, not a limited mode. It carries the full onchain venue set with no verification. Optional verification unlocks the regulated lane's centralized venues, which their own obligations require to serve identified counterparties. Same interface, same signed orders, deeper books.

The result is liquidity tiering rather than binary compliance. Verified users get deeper fills on size, and venues get a compliant counterparty pool without building distribution. One account spans both worlds, with the boundary placed where each venue's rules put it. The design goal is **one verification, whole lane**, a network-level credential accepted by regulated venues rather than KYC repeated per venue. Identity data stays with the verification provider and each venue's own onboarding. Pennysia orchestrates. It does not hold documents.

Open design decision. Credential mechanics: network-level attestation consumed by venues vs. per-venue onboarding with synced status. The first is the intent; venue acceptance governs which ships.

Eligibility applies per venue and per quote. Each regulated venue serves licensed jurisdictions, so some users will find the lane partially or wholly unavailable. The mechanism already runs in production as mechanical eligibility gates, formalized here as a lane rather than a switch. The limits are equally plain. Regulated-lane depth depends on venue onboarding and launches narrow, venue-rail settlement carries the bounded counterparty exposure in Section 7.4, and Pennysia provides no legal or tax determinations, though venue rules are enforced mechanically and displayed plainly.

9 Roadmap

Phases are ordered by dependency, not date. Each ships a working product (Table 3).

Table 3: Rollout phases: ordered by dependency, each shipping a working product with a concrete exit criterion.

Phase	Ships	Exit when
0. Prototype *(Ethereum mainnet)*	Parallel venue competition, Settlement with atomic fallback, WAP floor, four audit toolchains, public docs/SDK/status surface	The core loop (broadcast, compare, protect, execute) runs in production
1. Account + multi-chain spot	Embedded wallets (Web2 login → non-custodial EVM and Solana addresses), external-wallet parity, quote market across major EVM L2s, Solana, BNB Chain; signature-first same-chain fills	A user with only a Web2 login and USDC executes a same-chain swap: no seed phrase, no native gas token
2. Chainless execution	Cross-chain parity: intent rails under the same one-signature flow, unified balance, Settlement on major EVM networks, fallback across route chains, deeper venue coverage	Any asset on any chain to any asset on any chain, in one signature, WAP floor, one status line
3. Regulated lane	Verification tier; first centralized venues join the quote market; fiat ramps connect through the same credential (cards, bank, mobile pay); venue rails reach production; jurisdiction-aware eligibility	A verified user's large order is won by a regulated venue, in an interface unverified users never notice missing
4. Network scale	New service verticals join the quote market (perpetuals, prediction markets, yield vaults); more chain families; public order-format spec; documented APIs/SDKs	Third-party fillers and integrators build on the network

This paper should be judged on four metrics: onboarding completion (account to first settled trade without a seed phrase or gas token), cross-chain fill quality versus the best single-chain alternative, venue count and share of fills won per lane, and fill-verification rate together with failure-mode frequency.

A The Ethereum Prototype

This appendix records the experimental deployment on Ethereum Mainnet, live since August 28, 2026,[1] on which the OLN's design is based.

- **Parallel venue competition.** 30 venues quote per swap across four service classes (Table 4), reaching thousands of underlying liquidity sources. 12,000+ tokens listed.
- **Settlement contract.** Open-source onchain router (`0x3Aad...795E`): single token pull, ordered venue fallback in one transaction, cross-chain intent routes, full revert on all-routes-fail.
- **Protections.** WAP floor, gas-aware ranking, atomic auto-fallback, unified Settlement approval, double-layer security, the antecedent of Section 7's three layers.
- **Fees live.** Surplus fee on outperformance (10% cap, hardcoded), flat per-order protocol fee, partner share on intent routes. Every fee folds into the signed quote.
- **Audit toolchain.** Slither (static analysis), Halmos (symbolic execution, 168 tests), Medusa (coverage-guided fuzzing, 223 tests), Certora (formal verification, 105 rules). Reports public in the Settlement repo.
- **Defensive invariants.** Reentrancy guards via EIP-1153 [11]; venue/allowance-target may not equal the contract. Refunds are baseline-relative. Owner keys cannot raise the surplus cap.

Table 4: The prototype's 30-venue coverage by service class.

Class	Venues
DEXs & aggregators	onchain spot venues and meta-aggregators
Intent venues	same-chain and cross-chain intent networks
RFQ makers	quote-driven market makers
Privacy swaps	no-account, deposit-address providers; the ancestor of venue rails

B Order Format

A preliminary sketch of the order payload, the unit every venue competes to fill, signed as EIP-712 typed data on EVM [7]. Field-level spec publishes when the format stabilizes; this shows intent, not wire format.

```
OLN-Order {
  version:      uint16       // format version
  signer:       address      // embedded or external wallet
  input: {
    chain:      chainId      // origin chain
    token:      address      // asset sold
    amount:     uint256      // exact input
  }
  output: {
    chain:      chainId      // destination chain
    token:      address      // asset bought
    minAmount:  uint256      // WAP floor - worst acceptable output
  }
  lanes:        bitmask      // PERMISSIONLESS | REGULATED
  recipient:    address      // output delivery address
  deadline:     timestamp    // order expiry
  nonce:        uint256      // replay protection + uniqueness
  signature:    bytes        // EIP-712 typed-data (EVM) / equiv. (Solana)
}
```

`minAmount` **is the WAP**, the worst-case output the user signed, enforced on every rail. `lanes` scopes which venues may fill, and `REGULATED` requires a verified user. Fillers read the same object on every rail: take `input` on `input.chain`, deliver `output` on `output.chain`, at or above `minAmount`, before `deadline`.

[1] Live at https://app.pennysia.com; launch announcement at https://x.com/PennysiaLabs/status/2093384908354048180; contracts and audits at https://github.com/Pennysia.

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